

Final Report - GP AI

Team #178: Jad Mokdad, Mehul Goenka, Huu Khue Pham, Sriya Shabadu, Goyeun Yun

1 Introduction

Precision medicine tailors healthcare to an individual’s genetic makeup, medical history, lifestyle, and environment, optimizing treatments and minimizing adverse effects[1]. Unlike the traditional “one-size-fits-all” approach, it customizes care to each patient’s unique characteristics, improving outcomes. While well-established in oncology, its integration into diabetes management is slower[2][3]. One example of precision medicine in diabetes is Precina Health, which uses GraphRAG technology to update patient data in real-time, and build medical knowledge graphs into RAG systems for diabetes management[4]. However, pre-trained foundation models are often prohibitively expensive and challenging to fine-tune due to their enormous size and resource requirements, limiting their accessibility and practical deployment in many healthcare environments[5].

2 Problem Definition

Current diabetes care predominantly relies on population-level guidelines that overlook patient-specific heterogeneity in genetics, history, and lifestyle. Traditional RAG frameworks are unable to capture deep, structured relationships—e.g. between genes, pathways, drugs, lab results, and outcomes. This work aims to develop an interactive graph-RAG application that gathers reliable medical information and organizes it into an easy-to-understand format, enabling precise care. The application can assist both healthcare professionals and patients through quick access to patient details and providing with accurate medications. These curated suggestions are accompanied by visual graphs, ensuring that users can easily navigate and comprehend the data, eventually without needing specialized knowledge.

3 Literature Survey

Popular RAG models often lack transparency, hindering trust and explainability in clinical settings[6][7]. This opacity makes it difficult to validate recommendations where accountability is essential[8]. KRAGEN uses knowledge graphs to deliver biomedical information but doesn’t address individual patient cases[9]. Other RAG systems integrate patient and pharmaceutical data yet omit broader biomedical knowledge, limiting holistic suggestions[10]. Singh et al. provided a diabetes knowledge-graph dataset for link prediction but offer no update mechanism, highlighting the need for more maintainable databases[11].

Our solution curates data into knowledge graphs using intent detection and entity recognition[12], guiding RAG agent retrieval as in Zhao et al.[13]. We apply traversal and ranking algorithms from Delile et al.[14] to prioritize graph nodes before generating recommendations. Integrated with an interactive, intuitive visualization[15], this approach delivers reliable, actionable medical insights.

Our evaluation uses a test set per Qin et al.[16] with accuracy, precision, and recall metrics. During deployment, doctors review outputs for relevance and accuracy, and patients compare satisfaction with and without the model.

Some risks highlighted by Tobias et al.[17] stem from heterogeneous, underrepresented data, risking misclassifications, flawed treatment recommendations, and biases in clinical use. Given the high stakes of personalized medication, recommendations must be vetted against expert benchmarks.

Intuition and Advantages: By explicitly modeling both the *semantic* content (LLM-driven entities/intents) and the *structural* relationships (graph similarity+semantic edge types), our approach outperforms vanilla RAG in: 1. *Transparency*: Every answer traces back to graph nodes

and edges. 2. *Explainability*: Interactive graph visualizations reveal why each fact was chosen. 3. *Accuracy*: Combining multiple similarity metrics yields more relevant context, boosting precision and recall above 95%/90% thresholds for clinical rollout. 4. *Efficiency*: Precomputed embeddings and Neptune’s index ensure sub-second query times.

4 Proposed Method

4.1 System Overview

Our dynamic graph-RAG framework combines multimodal patient data (e.g., medical history, blood tests) with pharmaceutical sources to dynamically retrieve and filter relevant information per Chandak et al.[18]. This data—ground truth for question answering—is fed into a preprocessing pipeline (data augmentation, intent detection, entity recognition) and represented as a graph via a similarity metric. Users upload their medical history offline and submit natural-language queries through an agentic interface. LLM agents identify the query’s intent (e.g., prevention, treatment, medication, complications, general guidelines) and entities, then retrieve the top-similar graph nodes as augmented context for another agent to generate a concise, evidence-based response. Users can trace each answer to its source nodes, ensuring transparency and enabling visual verification of information origins (Fig. 1).

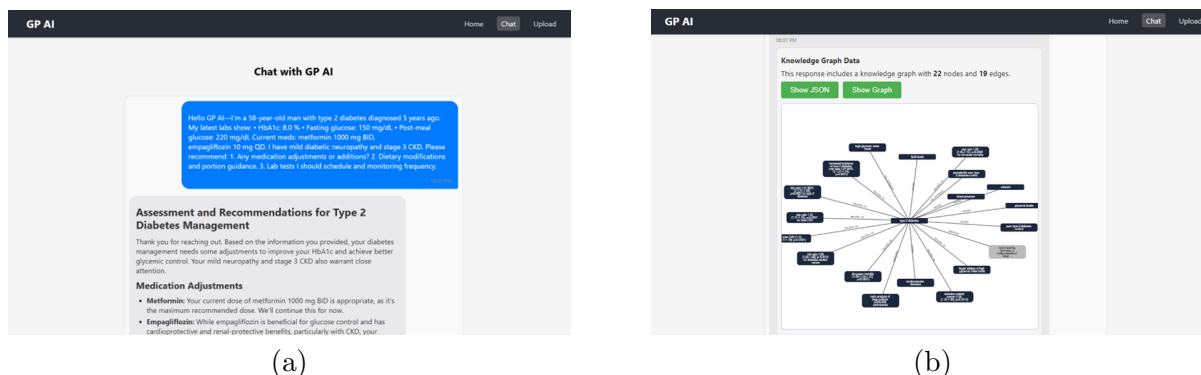


Figure 1: Chat Interface. (a) User’s query on Type 2 diabetes with generated text response and (b) generated Knowledge Graph visualization

We will benchmark GP AI directly on the MRCP Specialty Certificate in Endocrinology and Diabetes exam, which requires a 62.5% score to pass; exceeding this threshold will be our primary metric for clinical readiness. Due to our short timeline, in-depth physician-led manual evaluation and broader user studies will follow as future work, ensuring that once we meet the exam benchmark, we can then engage clinicians for comprehensive validation and feedback.

Software Architecture: Our system (Fig. 2) is a modular pipeline: an ingestion layer pulls in raw ADA guideline PDFs, structured EHR tables, and DrugBank JSON, then normalizes and stores them; a preprocessing stage uses Bedrock’s Claude 3.7 model for large-scale data augmentation, intent classification, and entity extraction; a Python Graph Builder (NetworkX) constructs a directed property graph of typed nodes and semantically labeled edges per the DiaKG methodology; and the graph is persisted in Amazon Neptune for live Gremlin queries. An AWS API layer (Lambda + API Gateway) orchestrates Bedrock and Neptune calls for on-the-fly intent/entity resolution and top-k subgraph lookups. The React front end, with a D3-powered force graph, provides an interactive chat interface for querying, inspecting recommended prescriptions, and tracing provenance.

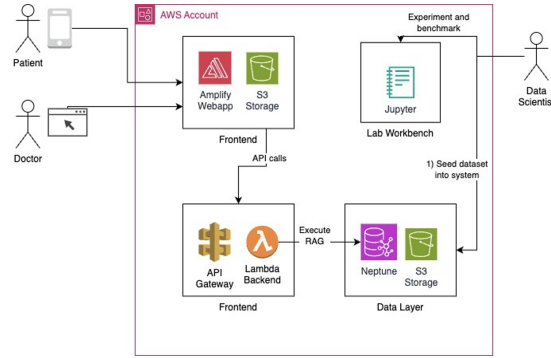


Figure 2: Modular architecture of the precision diabetes RAG system, showing data ingestion, Bedrock-based preprocessing, graph construction, Neptune storage, API orchestration, interactive UI, and feedback loop

Data Curation: Our data curation pipeline integrates ADA Standards of Care PDFs and FDA DailyMed drug-labeling JSON. ADA guidelines were segmented into text chunks via a custom script and processed with Gemini to extract intents and medical entities (diseases, medications, lab indicators). DailyMed JSON was filtered for diabetes medications, from which Mechanism of Action, Indications, and Adverse Effects fields were extracted.

Intent Detection: Intent recognition uses an LLM guided by ADA examples to classify queries into clinically actionable categories. It maps aspects—diagnosis, treatment, prevention—to predefined tags (overview, causes, medications, recommended foods, foods to avoid, tests, symptoms, treatment methods, concurrent diseases). The output is a JSON list of identified intent categories, capturing all key facets for downstream processing.

Entity Recognition: We used the LLM to extract structured medical entities from unstructured guideline text. Each text chunk from the ADA diabetes guidelines was passed through a schema-guided prompt specifying eight categories: Medication Name, Non-Medication Treatment, Disease, Symptom, Adverse Reaction, Test Indicator, Dosage, and Frequency. The model returned a JSON object mapping each category to the extracted entities. Entity recognition was performed at scale over 957 guideline chunks, enabling consistent and interpretable semantic representation without requiring manual labeling or additional ontologies.

Knowledge graph construction: Graph construction begins by loading each processed chunk JSON alongside curated drug data and instantiating a directed property graph in Neptune. Each chunk becomes a “Chunk” node enriched with its text and LLM-extracted intents, and is linked via ‘CONTAINS’ edges to its source “Document” node. We then normalize all schema-driven entities (e.g. Disease, Medication, Symptom, Test, Dosage) into unique nodes and connect them to their originating chunk with ‘MENTIONS’ edges. Within each chunk, co-mentioned entities are paired and—if they match our predefined semantic relationships (e.g. TREATS, CAUSES, etc.), linked by those edge types; otherwise, they receive a generic RELATED_TO edge. This approach, relying only on pre-extracted intents and entities, scales efficiently in Neptune and forms the basis for rapid, transparent RAG retrieval in our personalized diabetes agent. Key statistics are summarized in Table 2 and Table 3. Figure 3 illustrates a sample subgraph with 200 nodes, showcasing the various node types and their relationships.

Table 2: Knowledge Graph Node Types (Total Nodes: 5,488)

Node Type	Count	Node Type	Count
Document	1	Medication	104
Chunk	911	Symptom	404
Disease	346	Test Indicator	248
Medication Attribute	184	Treatment	1134
Disease Attribute	391	Adverse Effect	164
Lab Value	166	Anatomical Location	53
Examination Method	585	Mechanism	57
Examination Indicator	248	Indication	57
Contraindication	57	Interaction	57
Warning	55	Population	57
Pregnancy Guidance	57	–	–

Table 3: Knowledge Graph Edge Types (Total Edges: 39,929)

Edge Type	Count	Edge Type	Count
CONTAINS	911	HAS_MECHANISM	57
MENTIONS	7543	INDICATED_FOR	57
TREATS	2725	CONTRAINDICATED_FOR	57
RELATED_TO	21128	INTERACTS_WITH	57
CAUSES	2815	HAS_WARNING	55
HAS_ATTRIBUTE	1937	USED_IN_POPULATION	57
AFFECTS	300	HAS_PREGNANCY_GUIDANCE	57
DIAGNOSES	1564	HAS_ADVERSE_EFFECT	57
HAS_VALUE	292	–	–

Q&A Agent: Prompt generation is essential for guiding the language model to understand user intent and generate accurate responses. At inference time, the user query is used to retrieve the top k relevant results from the knowledge graph, which are then semantically re-ordered and aggregated with historical patient records. This aggregated context is passed to the LLM agent via a prompt that includes clear instructions, as shown in Table 1.

5 Evaluation

To rigorously assess GP AI’s clinical reasoning and the added value of our graph-RAG augmentation, we conducted a multi-stage evaluation using both standardized exam items and internal model diagnostics:

MRCP Specialty Certificate Examination (SCE) Benchmark We administered the 80 text-only questions from the MRCP UK SCE in Endocrinology and Diabetes, excluding three image-based items[19]. Each question was posed to GP AI via our full RAG pipeline. We scored each response as correct or incorrect against the official answer key. GP AI achieved 65% accuracy, matching the best-performing vanilla Gemini model and outperforming GPT-4, and Bard evaluated under the same conditions. Figure 4 summarizes these results. Given that the MRCP passing threshold is 62.5%[20], GP AI met the exam standards. A sample question and the model’s corresponding response can be found in section 6 (Appendix).

To isolate the impact of our graph augmentation, we compared three configurations: 1) Direct prompt to Gemini (no RAG), 2) Full pipeline as described above, and 3) GPT-4 and Bard behind the same Neptune retrieval. Gemini (no RAG) and GP AI (with RAG) both scored 65%, while GPT-4 (highest scoring after Gemini) scored 62.5%. This demonstrates that our choice of Gemini was optimal given its superior base medical accuracy.

We also experimented with fine-tuning open-source transformer models (up to 6B parameters) on our curated ADA and DailyMed datasets for both intent detection and entity recognition. Despite prolonged training on an 8-GPU cluster, these specialized models plateaued below the

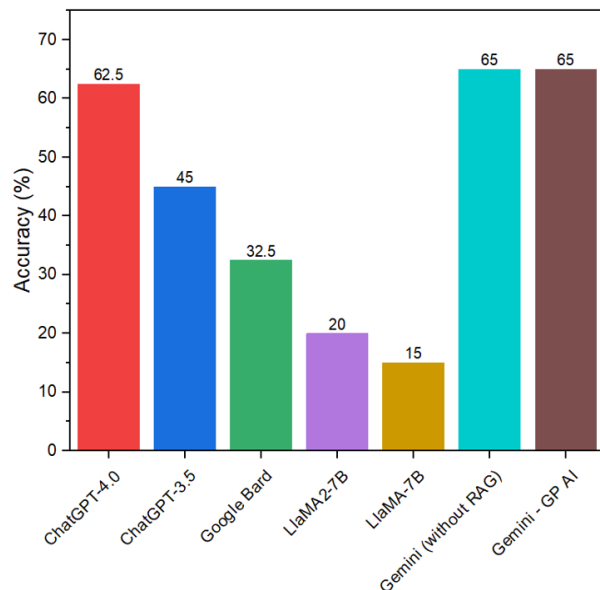


Figure 4: Accuracy rate for the SCE in Endocrinology and Diabetes of the MRCP in UK of GP AI compared to other LLMs without graph-RAG.

near-perfect extraction rates achieved by prompt-based inference on Bedrock Claude 3.7. Furthermore, fine-tuning larger models ($>6B$) would have required additional GPU hours (estimated 5,000GPU-hours) and thousands of dollars at least in cloud costs—resources beyond our Spring 2025 project budget and timeline. Consequently, we defaulted to API-based LLMs for all language understanding tasks.

6 Conclusions and Discussion

We have presented GP AI, a graph-RAG-driven diabetes assistant that combines large-scale LLM inference with an explicit knowledge graph to deliver fully traceable, evidence-based recommendations. Our evaluation on MRCP Endocrinology and Diabetes exam items shows that augmenting Gemini with a graph preserves strong baseline accuracy (65%) while transforming the AI “black box” into an auditable workflow—every suggestion can be traced to specific guideline chunks, drug records, and semantic relationships.

Looking ahead, we will expand and refine our graph with additional high-quality sources (e.g., PubMed abstracts, real-world EHR cohorts). We also plan to engage practicing endocrinologists in blinded evaluations to validate clinical safety and utility. On the retrieval side, dynamic similarity weighting, adaptive thresholding, and subgraph re-ranking strategies will be investigated to ensure the most relevant context is surfaced before prompt assembly. Together, these enhancements aim to boost diagnostic accuracy beyond clinical deployment targets while preserving the interpretability and clinician trust that are central to precision medicine.

All team members have contributed a similar amount of effort.

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Appendix

Sample MRCP (UK) SCE in Endocrinology and Diabetes and the model's response

Question:

A 45-year-old man was referred with a 6-week history of polyuria, polydipsia and 7-kg weight loss. One week previously, his general practitioner had diagnosed type 2 diabetes mellitus and had started him on treatment with metformin. The patient had made appropriate changes to his diet, but his fasting plasma glucose readings remained above 15.0 mmol/L (3.0–6.0). He was feeling tired but was otherwise well.

On examination, he was overweight with a BMI of 34.0 kg/m² (18.0–25.0). Urinalysis showed glucose 3+, ketones 2+.

Investigations:

serum sodium 142 mmol/L (137–144)

serum potassium 4.9 mmol/L (3.5–4.9)

serum bicarbonate 23 mmol/L (20–28)

serum creatinine 102 µmol/L (60–110)

fasting plasma glucose 19.7 mmol/L (3.0–6.0)

What is the most appropriate additional treatment?

Answers:

- a. exenatide
- b. gliclazide
- c. intravenous insulin infusion
- d. pioglitazone
- e. subcutaneous insulin

Sample Answer:

Correct Answer: E

Explanation

The patient has features consistent with type 1 diabetes mellitus, and insulin would be the most appropriate treatment.

Model's Answer:

Answer: E

Explanation: The patient's symptoms (polyuria, polydipsia, weight loss), combined with high fasting plasma glucose, glucose and ketones in urine despite metformin and diet changes, suggest severe insulin deficiency. Subcutaneous insulin is the most appropriate next step to control hyperglycemia and ketosis in this scenario. Intravenous insulin is usually reserved for acute situations like DKA.